

Email Setup Guide

This guide explains how to set up email functionality for Solo Ecommerce in development and production.

Quick Start (Development)

Option 1: Mailpit (Recommended)

Mailpit is a lightweight email testing tool that runs locally without Docker.

Windows Setup

1. Download Mailpit

- Go to: <https://github.com/axllent/mailpit/releases>
- Download `mailpit-windows-amd64.zip` (or the arm64 version if you have an ARM processor)
- Extract `mailpit.exe` to a convenient location (e.g., `C:\Tools\mailpit\`)

2. Run Mailpit

```
# Navigate to where you extracted mailpit
cd C:\Tools\mailpit

# Run it
.\mailpit.exe
```

Or add it to your PATH and run from anywhere:

```
mailpit
```

3. Access the Web UI

- Open: <http://localhost:8025>
- All emails sent by the application will appear here

4. Configure Backend

In `backend/.env` :

```
EMAIL_MODE=mailpit
SMTP_HOST=localhost
SMTP_PORT=1025
SMTP_FROM="Solo <no-reply@solo.local>"
```

5. Test It

- Register a new account in the app
- Check <http://localhost:8025> for the welcome/verification email

Mac/Linux Setup

```
# macOS with Homebrew
brew install mailpit

# Linux (download from releases)
wget https://github.com/axllent/mailpit/releases/latest/download/mailpit-linux-amd64.t
tar -xzf mailpit-linux-amd64.tar.gz
./mailpit
```

Option 2: Console Mode

If you don't want to run Mailpit, you can log emails to the console instead.

In `backend/.env`:

```
EMAIL_MODE=console
```

Emails will be printed to the backend terminal output:

```
=====
🔗 EMAIL (Console Mode)
=====
From: Solo <no-reply@solo.local>
To: user@example.com
Subject: Welcome to Solo! Please verify your email 📧
=====
[Email content here]
=====
```

Option 3: Docker (MailHog or Mailpit)

If you have Docker installed:

```
# docker-compose.yml
version: '3.8'
services:
  mailpit:
    image: axllent/mailpit
    container_name: solo-mailpit
    ports:
      - "1025:1025" # SMTP server
      - "8025:8025" # Web UI
    restart: unless-stopped
```

```
docker compose up -d mailpit
```

EMAIL_MODE Options

Mode	Description	Use Case
mailpit	Connects to localhost:1025, falls back to console if unavailable	Development with Mailpit
console	Logs emails to terminal only	Quick testing, CI/CD
smtp	Uses configured SMTP credentials	Production

Production Setup

For production, use a real SMTP service like SendGrid, AWS SES, or Postmark.

Example: SendGrid

```
APP_ENV=production
EMAIL_MODE=smtp
SMTP_HOST=smtp.sendgrid.net
SMTP_PORT=587
SMTP_USER=apikey
SMTP_PASS=your-sendgrid-api-key
SMTP_SECURE=false
SMTP_FROM="Solo <noreply@yourdomain.com>"
```

Example: AWS SES

```
APP_ENV=production
EMAIL_MODE=smtp
SMTP_HOST=email-smtp.us-east-1.amazonaws.com
SMTP_PORT=587
SMTP_USER=your-ses-smtp-username
SMTP_PASS=your-ses-smtp-password
SMTP_SECURE=false
SMTP_FROM="Solo <noreply@yourdomain.com>"
```

Example: Gmail (Not recommended for production)

```
EMAIL_MODE=smtp
SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_USER=your-email@gmail.com
SMTP_PASS=your-app-password # Not your regular password!
SMTP_SECURE=false
SMTP_FROM="Solo <your-email@gmail.com>"
```

⚠ For Gmail, you need to:

1. Enable 2-Factor Authentication
2. Create an "App Password" at <https://myaccount.google.com/apppasswords>
3. Use the App Password, not your regular password

Email Features

Currently Implemented

1. **Welcome + Verification Email** (on registration)

- Sent automatically when a user registers
- Contains a "Verify My Email" button
- Token expires in 1 hour

2. **Password Reset Email**

- Sent when user requests password reset
- Contains a "Reset Password" button
- Token expires in 1 hour

Endpoints

Endpoint	Method	Description
<code>/auth/register</code>	POST	Sends welcome/verification email
<code>/auth/verify-email</code>	POST	Verifies email with token
<code>/auth/resend-verification</code>	POST	Resends verification email
<code>/auth/forgot-password</code>	POST	Sends password reset email
<code>/auth/reset-password</code>	POST	Resets password with token

Troubleshooting

"Mailpit not available" warning

This appears when Mailpit isn't running. Either:

1. Start Mailpit: `mailpit.exe`
2. Switch to console mode: `EMAIL_MODE=console`

Emails not appearing in Mailpit

1. Check Mailpit is running on port 1025
2. Check backend logs for email send errors
3. Verify SMTP_HOST and SMTP_PORT in `.env`

Console mode not showing emails

Check the backend terminal output - emails are logged with the `[EmailService]` prefix.

Development Tips

View verification tokens in response

In development (`APP_ENV=development`), API responses include tokens for testing:

```
{
  "message": "Registration successful!",
  "verificationToken": "abc123...", // Only in development
  "user": { ... }
}
```

Quick verification without email

1. Register a user
 2. Copy the `verificationToken` from the API response
 3. Navigate to: `http://localhost:5000/verify-email?token=YOUR_TOKEN`
-

Security Notes

- Tokens are never stored in plain text (SHA256 hashed)
- Tokens expire after 1 hour
- Tokens can only be used once
- Token values are only returned in API responses when `APP_ENV=development`
- Always use HTTPS in production